Math Kangaroo Practice

Grade 3-4 · 20 questions · harder · seed 83478

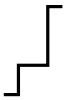
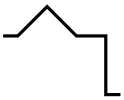
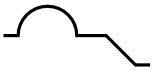
Part A: Each question is worth 3 points

1. Nicolas, Chloe, Ravi, Zoe and Sofia were born in different months: August, October, July, June and January. Nicolas was born in a month that starts with J. Sofia has a birthday in summer. Chloe was born in a month that starts with J. Chloe was not born in June. Sofia was born in a month that starts with J. Chloe and Ravi both have a birthday in summer.
Who was born in January?

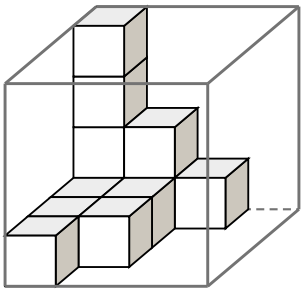
A. Ravi ***B.*** Chloe ***C.*** Sofia
D. Nicolas ***E.*** Impossible to determine

2. This is the outline of a town seen from the river at sunset.

Which piece **cannot** be part of the outline?

A. ***B.***  ***C.***  ***D.***  ***E.***

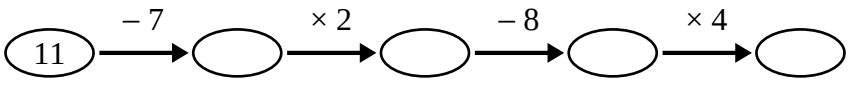
3. Leo has a glass aquarium shaped like a cube, 4 dm long, 4 dm wide and 4 dm tall. Leo has started to put cubes with edges of 1 dm into it, as in the picture. Every cube stands on the bottom or on another cube.



At most how many more such cubes can Leo put into the aquarium?

A. 52 ***B.*** 51 ***C.*** 45 ***D.*** 40 ***E.*** 12

4. Follow the arrows and do each calculation in turn.



Which number comes out at the end?

A. 11 ***B.*** 3 ***C.*** 2 ***D.*** 1 ***E.*** 0

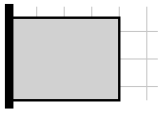
Part B: Each question is worth 4 points

5. A bar code is made of 17 black and white bars taking turns, with a black bar at each end. Each black bar is wide or narrow. There are 5 more white bars than wide black bars.

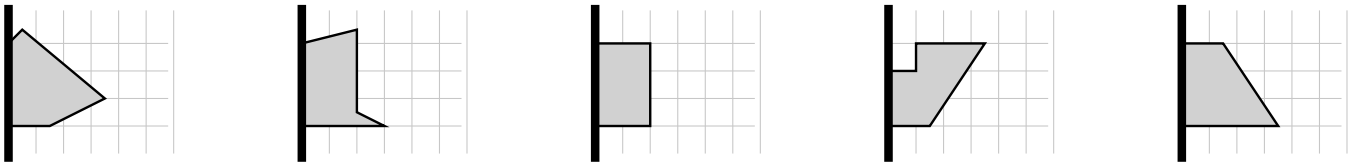
How many wide black bars are there?

A. 0 ***B.*** 1 ***C.*** 2 ***D.*** 3 ***E.*** 6

6. A rectangular flag flies on a pole in front of a wall, and the sun throws its shadow onto the wall. The wind turns and folds the flag, but the side sewn to the pole always stays on the pole. Mia draws the shadow at five different moments. Flat, the flag is 4 squares wide and 3 squares tall (see the picture). Some of the drawings are wrong.



the flag, flat



A B C D E

How many of the five drawings can be right?

A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

7. What is the biggest number smaller than 3018 whose sum of digits is the same as for 3018?

A. 3083 ***B.*** 3027 ***C.*** 3017 ***D.*** 3009 ***E.*** 2910

8. Basil was born on July 29, 2007 and Peter on August 28, 2007.
By how many days is Basil older than Peter?

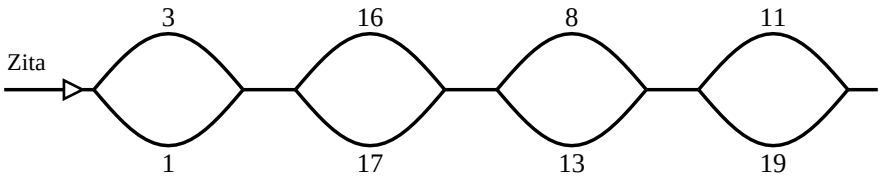
A. 37 ***B.*** 33 ***C.*** 31 ***D.*** 30 ***E.*** 1

9. The pages in a book are numbered from 1 to 148.
How many times is the digit 2 written?

A. 20 ***B.*** 34 ***C.*** 35 ***D.*** 36 ***E.*** 43

Part C: Each question is worth 5 points

10. Zita walks along the road from left to right and adds up the numbers along the way. The total is 36.



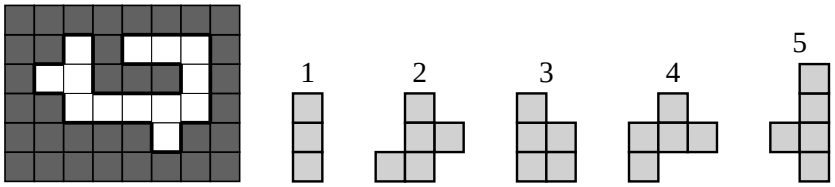
Which number did Zita see at the second fork?

A. 8 ***B.*** 13 ***C.*** 16 ***D.*** 17 ***E.*** 19

11. Mia types the numbers from 1 to 100 on a computer, with a comma between two numbers and no spaces: 1,2,3,...,100.
How many keys does Mia press (each digit and each comma is one key)?

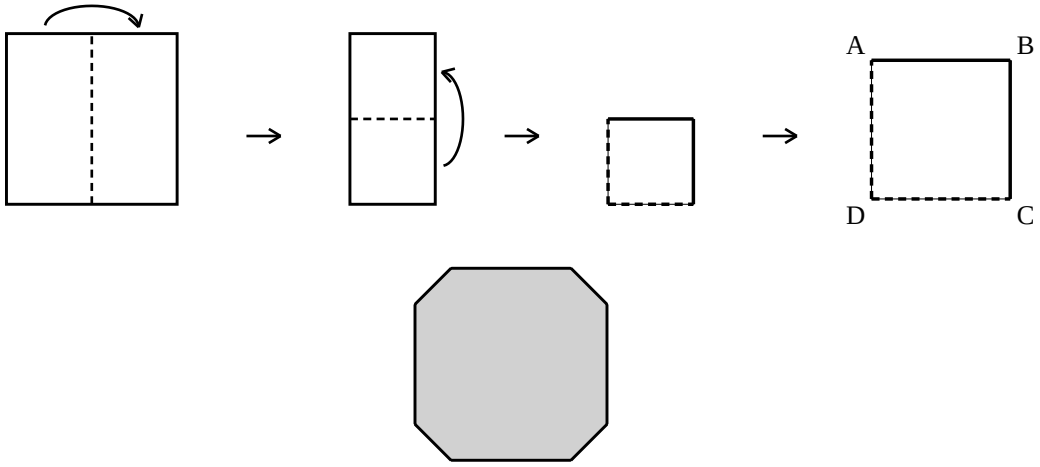
A. 200 ***B.*** 291 ***C.*** 292 ***D.*** 300 ***E.*** 350

12. Which three of these pieces can be used to cover the empty area exactly? The pieces may be turned and flipped over.



A. 1+2+3 ***B.*** 1+2+4 ***C.*** 1+2+5 ***D.*** 1+3+4 ***E.*** 2+3+5

13. A square piece of paper is folded in half, and in half again, so that it is a small square (see the picture: the dashed sides of the small square are the folds). One corner of the small square was cut off, and after unfolding the paper looks like the bottom picture.



Which corner was cut off?

A. corner A ***B.*** corner B ***C.*** corner C ***D.*** corner D ***E.*** it cannot be made with one cut

14. A big cube made of small cubes is painted on the outside and taken apart. Exactly 24 small cubes have 2 painted faces.
How many small cubes was the big cube made of?

A. 8 ***B.*** 48 ***C.*** 64 ***D.*** 96 ***E.*** 125

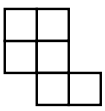
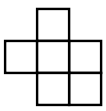
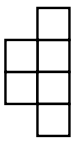
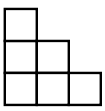
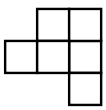
15. Five children, Emma, Grace, Iris, Hugo and David, sit in a row of 5 chairs at the cinema. Emma and Grace are best friends and want to sit next to each other.
In how many different ways can the children sit?

A. 4 ***B.*** 8 ***C.*** 24 ***D.*** 39 ***E.*** 48

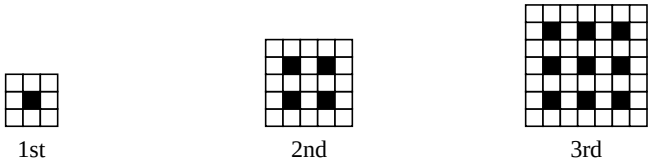
16. How many different letters are used to write the two words LEMONADE and TURTLE?

A. 11 ***B.*** 10 ***C.*** 9 ***D.*** 8 ***E.*** 7

17. Mia has two copies of the same piece. With which piece can she make a rectangle? Pieces may be turned and flipped over.

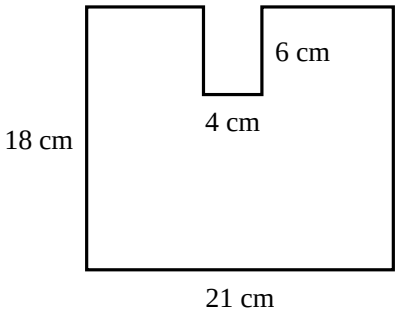
A.  ***B.***  ***C.***  ***D.***  ***E.*** 

18. The diagrams below are the 1st, 2nd and 3rd of a pattern. One diagram further on has exactly 225 white cells. Which one is it?



A. the 7th ***B.*** the 8th ***C.*** the 10th ***D.*** the 11th ***E.*** the 12th

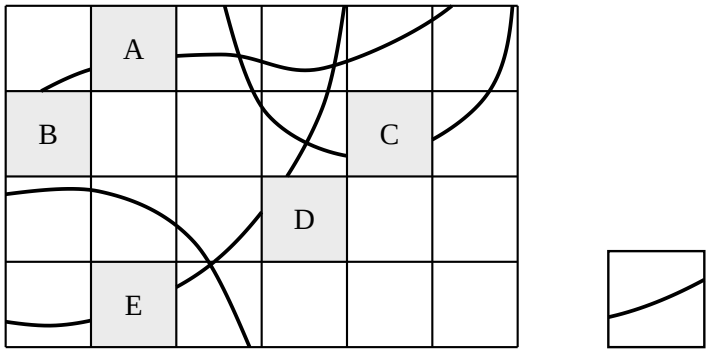
19. A 21 cm by 18 cm rectangle of cardboard has a rectangular notch 4 cm wide and 6 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 111 cm ***B.*** 106 cm ***C.*** 99 cm ***D.*** 98 cm ***E.*** 90 cm

20. Some paths go across a park. Five squares are missing, marked A to E. The square on the right is one of them (it is not turned).



Which gap does it fit into?

A. gap A ***B.*** gap B ***C.*** gap C ***D.*** gap D ***E.*** gap E

Answer Key

1.	**D**	**2.**	**A**	**3.**	**A**	**4.**	**E**	**5.**	**D**	**6.**	**D**
7.	**D**	**8.**	**D**	**9.**	**C**	**10.**	**C**	**11.**	**B**	**12.**	**B**
13.	**B**	**14.**	**C**	**15.**	**E**	**16.**	**B**	**17.**	**D**	**18.**	**B**
19.	**E**	**20.**	**E**								

Solutions

1. D (Nicolas) — The only possible matching: Nicolas – January; Chloe – July; Ravi – August; Zoe – October; Sofia – June. [2002 Q18 · who_has]

2. A — All the other pieces can be found in the line, but this one cannot: it has a round dome where the line has two small bumps. [2002 Q7 · cannot_belong]

3. A (52) — The aquarium holds 4 × 4 × 4 = 64 cubes. There are already 12 inside (count column by column, also the hidden ones). 64 – 12 = 52. [2007 Q14 · how_many_more]

4. E (0) — 11 – 7 = 4, 4 × 2 = 8, 8 – 8 = 0, 0 × 4 = 0. [2007 Q5 · forwards]

5. D (3) — There are 9 black and 8 white bars. Wide: 8 – 5 = 3. [2003 Q17 · wide_bars]

6. D (4) — A drawing is wrong if part of the pole edge is bare or if it reaches more than 4 squares from the pole. Wrong: D. So 4 can be right. [2002 Q15 · count_possible]

7. D (3009) — 3 + 0 + 1 + 8 = 12. Counting down from 3018, the first number with digit sum 12 is 3009. [2007 Q4 · before_same_sum]

8. D (30) — July has 31 days, so from July 29 to the end of the month is 2 days, then 28 more days to August 28, 2007: 30 days. [2007 Q12 · days_between]

9. C (35) — As the last digit: 15 times; as the tens digit: 20 times. Total 35. [2003 Q16 · count_one_digit]

10. C (16) — Only one route gives 36: 1 + 16 + 8 + 11 = 36. At the second fork: 16. [2007 Q1 · which_branch]

11. B (291) — Digits: 9 × 1 + 90 × 2 + 1 × 3 = 192. Commas: one fewer than the numbers, 99. 192 + 99 = 291. [2007 Q23 · with_commas]

12. B (1+2+4) — The empty area has 13 squares, so the three sizes must add up to 13. Only 1 + 2 + 4 also fit together in the right shape. [2003 Q18 · three_pieces]

13. B (corner B) — The design shows all four corners cut off. That comes from cutting corner B (the real paper corner, away from the folds). [2007 Q24 · which_corner]

14. C (64) — A cube has 12 edges, so each edge has 24 ÷ 12 = 2 cubes between its corners. With the 2 corners, an edge is 4 cubes long, and the cube has 4 × 4 × 4 = 64. [2007 Q17 · how_big]

15. E (48) — Think of Emma and Grace as one block: the block and the other 3 make 4 things to order, 4 × 3 × 2 × 1 = 24 ways. Inside the block Emma and Grace can swap, so 2 × 24 = 48. [2002 Q22 · side_by_side]

16. B (10) — LEMONADE uses 7 different letters and TURTLE uses 5. E and L are in both, so count them once: 7 + 5 – 2 = 10. [2007 Q3 · letter_cards]

17. D — Two copies have 12 squares, so the rectangle is 1 × 12 or 2 × 6 or 3 × 4. Only two copies of piece D fit together like that. [2007 Q8 · two_copies]

18. B (the 8th) — Try the diagrams one by one: the 8th is 17 × 17 = 289 cells with 64 black, so 225 white. [2007 Q21 · which_diagram]

19. E (90 cm) — The bottom of the notch (4 cm) replaces the 4 cm taken out of the top edge, and the two sides of the notch add 2 × 6 cm. Perimeter = 2 × (21 + 18) + 2 × 6 = 90 cm. [2002 Q6 · notch]

20. E (gap E) — The square has top: no line, right: 1 line, bottom: no line, left: 1 line. Only gap E has lines reaching it at exactly those places. [2002 Q1 · which_gap]